Hardik .K. Singh

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Education

Vellore Institute of Technology, Oct 2022 – July 2026
B.Tech in Computer Science and Engineering Specialization in Health Informatics **(75%)**

Skills

Data Analysis: EDA, ETL, Python, SQL, Pandas, NumPy, Data Cleaning.
ML/AI: n8n, Vector search, Vector embeddings, RAG (Retrieval Augmented Generation).
Web Development: React.js, Java, HTML/CSS, JavaScript.
Tools: Git, Power BI, Tableau, VS Code.

Projects

Customer Shopping Behaviour Analytics & AI-Powered Insight Automation
Python (Pandas),SQL,PostgreSQL,Power BI, n8n,OpenAI API

- Designed an end-to-end retail analytics workflow — performed EDA and data cleaning in Python (Pandas), engineered advanced PostgreSQL queries for revenue segmentation and discount evaluation, and built an interactive Power BI dashboard tracking key retail KPIs, reducing manual reporting time by 60%.
- Built an n8n automation pipeline that ingests cleaned sales data on a scheduled trigger, routes it through OpenAI API to auto-generate weekly insight summaries (top segments, anomalies, revenue dips), and delivers findings to stakeholders via email — eliminating recurring manual reporting with zero manual intervention.

Healthcare Management System
React.js, Node.js, Express.js, MongoDB

- Developed a full-stack healthcare application with appointment booking, patient registration, and RESTful APIs — enabling structured data collection and storage that supports downstream data extraction and analysis workflows.
- Designed a responsive frontend and deployed on Vercel achieving 99% uptime; the system's structured MongoDB schema and API layer enabled clean, queryable datasets for reporting and business intelligence use cases.

AI-Powered Email Triage & Automated Insight Routing System
n8n,OpenAI API,Gmail API,Google Sheets

- Developed and trained CNN-based gesture recognition model to classify 1,000+ sign language gestures with 88% accuracy, implementing data augmentation and hyperparameter tuning to optimize performance across diverse lighting conditions and hand positions.
- Built end-to-end ML pipeline including dataset preprocessing, model training, and validation with 500+ user interactions, applying transfer learning techniques to reduce inference time by 50% while maintaining high accuracy for real-time video processing.

Awards & Recognition

Health Hackathon Participant | Johns Hopkins University, USA Feb 2023

- Selected among top participants to develop AI-powered healthcare solutions addressing real-world medical.

Second Position, IEEE Q-RiOSITY Competition | VIT Bhopal July 2022

- Secured 2nd rank among 100+ participants demonstrating engineering knowledge and rapid problem-solving.

Certifications

- AI Foundation Course – Jio Institute